

Data infrastructure studies on an unequal planet

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Abstract

In this article, I take the case of data centers as a powerful tool and infrastructure of multinational digital capitalism, analyzing the ways in which understanding these and other data infrastructures through their energy frameworks allows us to theorize the implications of planetary environmental impacts of digital data for contemporary subjects beyond individual data technologies themselves. This is especially true in data centers' function as energy vacuums and in their carbon and extractive footprints and other environmental externalities. I demonstrate that data centers organize an assemblage of environmental relations whose operations reproduce uneven systems of capitalism enacted through energy and environmental politics. While this article is by no means comprehensive, and by necessity must be selective in its engagement with key texts in a number of overlapping fields, it broadly draws from media studies, geographical, and sociological approaches to data infrastructures to unravel the entanglements of digital systems and the environment. Data centers and their energy connections represent multivalent sites and indications into the global supply chain of data infrastructure, and their extractive dynamic as networked infrastructure fundamentally changes how we need to see their impacts and the impacts of datafication more broadly.

Keywords

Data centers, data infrastructures, data colonialism, energy, extractivism, environmental media

In this article, I take the case of data centers as a powerful tool and infrastructure of multinational digital capitalism. Drawing inspiration from environmentalists Anil Agarwal and Sunita Narain's concept of "environmental colonialism," I demonstrate that data centers organize an assemblage of environmental relations whose operations reproduce uneven systems of capitalism enacted through energy and environmental politics. Data centers are energy vacuums (Bresnihan and Brodie, 2021) and have extensive carbon and extractive footprints and other environmental externalities (see Pasek, 2019). Engaging key texts across media studies, geography, and sociology, the article sits within a proliferating landscape of "data center studies" that has taken off in the past several years (see Hogan, 2021; Johnson, 2019; Velkova, 2016; Vonderau, 2019). As a critical scholar of data centers and specifically the cultural, social, and environmental politics that arise at sites of their development (see Brodie, 2020a, 2020b; Bresnihan and Brodie, 2021, 2023; Brodie and Velkova, 2021), this article coalesces this work into a theory of the uneven planetary implications of data infrastructures. To do so, I use empirical research into the data center industry to support a review of secondary literature on digital data's expansive infrastructural networks and energetic ecologies.

Foundational data center studies have of course demonstrated their significant environmental impacts (Carruth, 2014; Cubitt et al., 2011). Recently, scholars have specifically analyzed the role of digital infrastructures within energy systems and so-called "green" transitions (see Bresnihan and Brodie, 2021, 2023; Lally et al., 2022; Ortar et al., 2023; Velkova, 2021) and their interactions with local economies and environmental politics (see Brodie, 2020b; Burrell, 2020; Hogan, 2015; Johnson, 2019; Vonderau, 2019). Their energy ecologies represent multivalent entries into the global supply chain of data infrastructure, which spans networked extraction sites, transportation, cables, atmospheres, geologies—underground, overground, airborne, and undersea processes

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shaped by histories of colonial capitalism and the uneven effects of climate crisis.

The article is also indebted to work in critical data studies which analyzes digital data's unequal social formations and functions, including essential research on discriminatory data (Chun, 2021; Eubanks, 2017; Noble, 2018), data rights/justice (Taylor, 2017), Indigenous data sovereignty (Reardon and Tallbear, 2012; Rodriguez-Lonebear, 2016), and "data from the south" (Amrute et al., 2022; Milan and Treré, 2019; Ricaurte, 2019). All of this work takes different and context-specific aim at the various forms of the "coloniality of data" that Nick Couldry and Ulises A. Mejias identify in their analysis of what they call "data colonialism" (2019a, 2019b). In doing so, this diverse work speaks to the various ways that the power structures of digital economies, from "AI colonialism" in the global south (Hao, 2022) to the surveillance of colonized and racialized people in the global north (Browne, 2015; Cormack and Kukutai, 2022), recreate and reinforce the technoscientific rationalities and epistemologies of a continually unequal world order (see Kwet, 2021). In alliance with these approaches, this article argues that dismantling the foundational premises on which the infrastructural backbones of capitalist data economies are formed is an essential precondition for truly *environmental* data justice. Sites of extraction and environmental violence supporting digital operations are central to the form and function of global data economies. As such, this article joins a growing literature surrounding the material dynamics of extractivism as embedded not only within the structures of "data extractivism," but the networks of resource-making and territorial dynamics through which this particular formation of global capitalism is pillaging the planet (see Lehedé, 2022c; Taffel, 2023).

While recognizing the crucial import of perspectives from the "majority world," my empirical examples will draw from contexts of uneven development in the global north, primarily Ireland, a "semi-peripheral" country hosting the highest concentration of data centers in Europe. Focusing on such a case, while also reviewing critical literature from global contexts, creates a position from which to observe the intensive and extensive dynamics of data infrastructural development and the planetary impacts of tech capital. Putting data center studies in conversation with critical data studies approaches to "data colonialism" and "data extractivism," this article considers how data centers act as material and discursive infrastructures within the maintenance of planetary inequities during simultaneous "datafication" and green energy transitions.

The article will thus make a theoretical contribution to data center studies by demonstrating the centrality of these digital infrastructures within expanding networks of extraction and the uneven development of "green" capitalism as it maps onto intensifying planetary imbalances. Drawing on 6 years of field and policy research in Ireland

on the global data center industry, its relations to energy systems, and its extractive activities and rationalities, this article argues that a more robustly geographical and infrastructural approach to data politics will be required to support anti-extractivist and environmental justice frameworks and movements, as data systems are increasingly intertwined in "green" transitions and emerging "smart" systems for managing them. I will begin by historicizing how climate responsibility drawn from global north-centric discourses flattens multiscalar and uneven geographical distribution of energy and resource use, with major implications for the use and location of data technologies and infrastructures. The article then analyzes the intensive and extensive dynamics of digital growth which map the geographical unevenness of datafication. Turning to theories of "data colonialism," to date the most comprehensive (and debated) conceptual framework for the simultaneous developmental and representational elements of datafication, the article then demonstrates that these concepts knock off the balance of the epistemological and the material, requiring more intensive focus on the infrastructural energies and resources that fuel data systems. The final section then draws on approaches to "data extractivism" to demonstrate how processes of resource making, and especially through the expansion of datafication-enabled renewable energy, finds a "natural" extractivist architecture in the form of the data center. The article concludes, then, with speculations about the scale of action that could set in motion the redistribution of responsibility for climate change and its adaptation without resorting to technological fixes, determinisms, and onshoring, presenting a modest contribution and call to action for data infrastructure studies on an unequal planet.

Infrastructure out of balance

In their 1991 paper "Global Warming in an Unequal World," Agarwal and Narain (2019) claim that emissions measurements across the global north and the global south are incomparable. Articulating what they call "environmental colonialism," they concisely detail the faulty mathematics of Eurocentric climate science and its colonial biases, emphasizing that sole responsibility for climate change lies in the laps of gas guzzling global north societies. In a scathing refutation of an influential World Resources Institute (WRI) Report, they argue "no effort has been made in WRI's report to separate out the 'survival emissions' of the poor, from the 'luxury emissions' of the rich. Just what kind of politics or morality is this which masquerades in the name of 'one worldism' and 'high minded internationalism'" (pp. 82–83)?

Although the overarching climate metrics may have changed (and gotten even more grim), the sentiment identified by Agarwal and Narain remains pertinent: that any attempt to flatten, or even *compare*, the environmental

responsibility of enormously powerful, wealthy nations or institutions and that of poor, racialized, or otherwise excluded or dispossessed communities, abandons an ethic of geopolitical critique that accounts for the unevenness of ongoing global structure(s) of colonialism. These politics remain today—after all, in 2021, Duke University researchers raised controversy when they announced a project proposing that “replacing wood burning stoves in sub-Saharan Africa with cleaner technology could offset carbon emissions of a mid-sized country” (Penn, 2021). This flattened view of climate responsibility—where survival emissions are discursively scaled-up and conflated with large-scale and population-scale carbon emissions—has been popular across social science approaches to environmental issues in the global north for decades. For example, geographer Fergus (1983), in a 1980s study of the use of firewood for household and community heating in “rural energy markets” in north-west Tanzania, concludes that small-scale and household energy uses of firewood and coal in this “remote” region is an oncoming ecological “crisis” to be remedied with large-scale hydropower. This neocolonial rubric, enacted by pastoral developmentalism and renewable energy, continues and intensifies in the present, under the guise of a sustainable “future” while activating destructive colonial relations of capital supposedly already relegated to the past. But perhaps most interesting to draw out, and to begin the process of updating Agarwal and Narain’s remarks to the contemporary situation of data and energy at a global scale, is Fergus’ proposed *solution*: investment in renewable energy infrastructure to heat homes, provide electric cooking utilities, and to reduce household carbon emissions. But who benefits? Who gets the contracts to build? Who administers the utility? How long until it becomes operational and usable? Whose houses get retrofitted? Whose ways of life are accounted for, or not? Who is held responsible, and who is made to shoulder costs and responsibility in whatever form?

The incommensurability of yearly wood-fired stove emissions with even one transatlantic flight on a commercial (not to mention private) airplane should alone be enough to raise eyebrows about this widespread, pervasive methodology of individualization, which rests on an ethic of “personal responsibility” as it affects global climate and environmental issues, ignoring existing imbalances of global systems. According to recent studies, the top 1% of global emitters worldwide produce 75 times more carbon emissions than the bottom 50% (Tandon, 2022). Certainly, individuals and even discrete communities have responsibilities, obligations, and relations as some critical Indigenous scholars frame it (see Kimmerer, 2002; Liboiron, 2021; Spice, 2018), to act toward a less harmful world, especially in the context of environmental (and land) practice and action. However, in the context of climate justice and the solutions presented, the mechanisms by which this responsibility enters the mainstream are overwhelmingly pushed by large corporations and states. These

powers inherit the “one-worldism” of global climate science centered on the undifferentiated, wealthy global north (see Young, 2021), wherein adaptation to anthropogenic climate change *caused* by colonial capitalism is imposed upon those still or historically colonized, for whom such existential catastrophe has been occurring for centuries (Whyte, 2018). “Green grabbing” and “green extractivism” in the global south and in Indigenous communities are promoted as essential interventions for climate adaptations, reproducing and intensifying uneven geographies of accumulation (see Bruna, 2022; Fairhead et al., 2012). There is no “one world,” no planetary politics on equal footing, if we take seriously structural and spatial imbalances maintained by supply chain capitalism, especially those being built and reproduced by the emergence of green and smart digital capitalism.

Today, as societies worldwide shift slowly to “renewable” forms of energy and the emerging infrastructural and technological systems these entail—including digitally enabled energy technologies like smart domestic electricity and responsive grids (see Levenda et al., 2015; Velkova et al., 2022)—new personal responsibilities arise to reduce carbon emissions and limit individual climate impacts, presenting new formations of existing inequalities that are intensified through digital data relations. These individualized assumptions can be seen in many global north-centered climate and environmental discourses, and impacts of Internet usage—particularly the energy use of individual digital practices like streaming and cloud storage in comparison to the astronomical use of multinational data centers—has been a pertinent recent example of this scalar imbalance (see Marks and Przedpeński, 2021). Data infrastructure of course entails an assemblage of infrastructural sites and networks including fiber optic cables, content distribution networks, and telecoms masts as well as data centers to function (see Parks and Starosielski, 2015). The geographical imbalances of digital data infrastructure thus also involve discussing who has and who controls access to broadband and bandwidth, especially in rural and marginalized communities (see Ali, 2021; Duarte, 2017; Randell-Moon and Hynes, 2022). You can stand on a beach in rural Mayo, in Ireland, meters from a transatlantic fiber optic cable owned by a multinational tech consortium—and still have no basic cell phone service. Nearby, a different consortium (including Facebook and Google) has just landed yet another cable, which crosses Clare Island seabed off the coast. Despite objections centered on concerns over fishery disruption and the islanders’ lack of reliable digital access, the cable has gone ahead, a high-speed and passively profitable digital infrastructure “connecting” the world by bypassing a “disconnected” region of rural Ireland. Inheriting in many ways the “one-worldism” of the environmental discourses and policies maligned by Agarwal and Narain, the imagination of data

infrastructures' planetary expansion—whether triumphalist (see Stephenson, 1996) or critical (see Edwards, 2013; Gabrys, 2016)—in large part fails to account for the unequal relations sustained by and within its *uneven* reach into the patchwork territories of the earth.

Crucial to any understanding of these imbalances is also essentially energetic. From the earliest experiments in distanced telecommunication, wired and wireless electronic information transfer at scale required vast amounts of energy (Thibault, 2014). Today, data centers, repositories of digital access and the site of collective consumer and business activity of streaming and “the cloud,” are energy hogs, demanding about 1% to 2% of global electricity supply (and much more in individual national contexts, including around 14% of the grid in Ireland in 2021, more than all rural Irish households put together (CSO, 2022)). In hot and dry places, they use disproportionate amounts of water for supplemental cooling (Hogan, 2015), not to mention the supply chains of the minerals and manufacturing of the technologies inside them, usually sourced and built far from where they are operating. They are usually owned by huge multinational corporations with immense spending power and influence over government economic policies. This scale thus extends beyond the often imposing (although typically nondescript) architectural structures, the “hyperscale” facilities usually built in suburban industrial and logistical parks encircling major cities. In these facilities, the digital activities undergirding contemporary life—from everyday social media use to public service provision—are computed, representing concentrated resource-intensive sites at which society-wide activities are supported and profited from private institutions administering the infrastructure and the software of its operation (see Maguire and Winthereik, 2021; O'Neill, 2018). But while an apparently necessary infrastructure, the basis of the data center business model is the uninterrupted growth of digital data (Jung, 2023) and the concentration of tasks, services, and computing power in their proprietary “cloud” servers, making the material footprints of digital and the attendant energy usage inextricable from the business interests of the companies increasingly involved in these diverse transformations in infrastructural provisioning.

As feminist media scholar Nancy Mauro-Flude argues in the context of Singapore's data center economy, visions of an undifferentiated landscape of endless datafication reproduces “the singular, global, monolingual Internet that acts as a homogenizing technology, persistently eradicating difference” (2023: 27). Digital growth looks different across global geographies, and responds to lumpy infrastructural affordances and environmental antagonisms in local jurisdictions. The concentration of data infrastructures like data centers also maps onto the “the presence of infrastructural conduits that date back to previous episodes of imperial expansion” (p. 27). In the uneven growth of digital

systems, the geographical imbalances of existing global circumstances are frequently intensified rather than rectified by their expansion, as new frontiers require new (and frequently violent) material efficiencies, especially in the form of energy.

Growth, intensification, and extractive efficiencies

At the heart of these data (and energy) relations is a technoscientific rationality that supplants the value epistemologies and scale of industrial modernity with the speed and perceived efficiency of globalization. These operational efficiencies are enabled by the much lumpier and distributed growth of digital infrastructures like data centers, which are centrally implicated within emerging infrastructural systems of increasingly paired “smart” and “green” technologies. Industry groups, for example, endeavor to make data centers *essential* rather than only *subject to* the governance of energy transitions to intermittent, atmospheric energy sources (see Bloomberg NEF and Eaton, 2021), whether through the deployment of internal sensing, the absorption of excess energy in their backup battery arrays, or timing their computing loads to be “grid responsive” to the greatest production of renewable energy compared to lower population-wide consumption (see Liu and Terzis, 2012; Radovanovic, 2020; Roach, 2022). In doing so, data centers are becoming energy silos, wherein the companies that run them are siphoning public energy, innovating their own efficiencies, and profiting off energy transitions.

In Ireland, this has come to the fore as multinationals were reportedly charged a fraction of what ordinary consumers paid for bulk electricity from providers during the 2022 to 2023 energy crisis (O'Halloran, 2023), meaning that data centers, which drew 14% of Ireland's energy daily in 2021 and are projected to draw 28% by 2031 (Eirgrid and SONI, 2022: 9), are getting this energy at a discount. Mechanisms like corporate power purchase agreements also secure low corporate pricing by guaranteeing set energy prices years in advance for companies, who are then able to promote their commitments to decarbonization by purchasing renewables like wind generated in rural Ireland (Bresnihan and Brodie, 2021). At the state level, these schemes are promoted as a primary strategy for building out renewable energy into the future, as these agreements “can reduce risks and save money and carbon for companies, and can also contribute to accelerating decarbonisation and reducing costs for the wider economy” (Government of Ireland, 2022: 7). According to the Commission for the Regulation of Utilities, “in an open, competitive market, it could not regulate the prices that each supplier charged a particular group” (O'Halloran, 2023). The Irish grid is still primarily powered by fossil

fuels like gas, but multinationals are able to secure contracts ostensibly supplying cheap “renewable” power over consumers under the guise of sustainability. Meanwhile, consumer-citizens endure state utility PSAs on TV and Instagram through the Government’s “Reduce Your Use” campaign, designed to encourage consumers to reduce their energy use to contribute to society-wide decarbonization and cut astronomically expensive household energy costs—which extend from electricity to solid fuels. Energy policy currently allows private tech companies to exploit the growing renewable energy market, while ordinary consumers face energy poverty and minimal public support.

These imbalances are experienced vividly on-the-ground, especially in rural Ireland, where heat is difficult to generate and maintain efficiently in often poorly insulated homes unconnected to public heating utilities or retrofitted for electric heating. Yet, any rural wind farm could be under contract to provide electricity directly to Amazon, Google, or Microsoft hundreds of miles away in Dublin. You can thus potentially live the shortest legal distance from a multinational wind farm, able to charge your laptop to use all of the different subscription services you need to access basic internet content for consumer and business activity (if you are lucky enough to have broadband coverage)—and still have to burn carbon fuels like turf, coal, or kerosene to heat your home. What is important at this point in the article is to understand that siloed corporate data/energy politics—wherein big tech companies control and profit from what is ostensibly a transition administered through public energy systems, albeit in response to global regulations with planetary climate aspirations—reproduce uneven impacts, accumulation, and development across infrastructural geographies, making data centers a crucial technology within a planetary system of environmental inequality.

This extends far beyond individual and even corporate-scale energy use and into how these systems are being transformed under changes in resources and infrastructural governance. As energy anthropologists Howe and Boyer (2016) articulate, “Failure to rethink an extractive model of energy production could likewise result in deepening geopolitical inequalities and lead, possibly, to a form of climatological imperialism in which the global South is tasked with rehabilitating the (much more historically contaminative) global North” (p. 218). An energy production model which is designed to deliver benefits directly to multinational industries far from sites of extraction appears familiar within Howe and Boyer’s description, and of course, in the form of wind in Ireland, its primary “green energy” resource, represents an atmospherically networked intensification of internally uneven geographical and infrastructural relations between its rural and urban regions (Breathnach, 1988; McCann, 2011).

Approaches to extractive data relations also identify the legacy of colonialism within contemporary systems—

whether as a result of infrastructural path dependence or simply “old-school” colonialism enacted via neoliberal economic restructuring. Each has its own way to deal with the intensive and extensive elements of data and energy supply chains, which must be brought together to understand the intersecting character of data infrastructural politics today. This article does not, by any stretch, “correct” the imbalances recognized, epistemologically and spatially, by Agarwal and Narain, especially in terms of re-centering the global south. My primary research context, after all, is a country in the global north, albeit with its own complicated colonial histories and tax haven status making it an important case study into the imbalances and dependences of global digital infrastructures (see Bresnihan and Brodie, 2023; McCabe, 2022; O’Boyle and Allen, 2021). But rigid global distinctions should not foreclose recognition of commonality and solidarities across uneven geographic contexts. For example, struggles around digital and infrastructure rights in urban contexts in Chile share important overlaps with struggles around energy and digital sovereignty in urban and rural contexts across places like Ireland and the Netherlands (see Lehuedé, 2022a; Rone, 2023a). The histories and experiences of climate change, and digital capitalism, are undoubtedly much different in Dublin than they are in Santiago, and unequal class and racial relations within and across these places, and their rural hinterlands, still map different complexities of climate vulnerability. But this does not mean that we should not think about the systemic relations between places navigating the structural violence of climate change and its adaptations through digital technology.

The perennial question is whether data infrastructures, the economies they support, and the capital they generate, will materially benefit local people in any equitable way, especially balanced against visible damages to the environment. Struggles around data infrastructures, like those around digital rights more broadly, are often ambivalent, ranging from local community and civil society support for the development of data centers to committed objection and protest on grounds of social and environmental justice (see Brodie, 2020b; Rone, 2023a, 2023b). These are infrastructural struggles necessarily tied into other resource frameworks around land, energy, and water. For example, in Ireland, data centers from companies like AWS, Facebook, Google, and Microsoft have outstripped the constraints of a grid unable to handle any more connections (see Sargent, 2022), and in Chile, engaged opposition has been somewhat successful in stopping extractive multinational data projects (Lehuedé, 2022c). But how do these struggles, formed around apparently isolated sites of multinational extractivism, scale outwards from localized infrastructural and resource constraints? And how do we account for the compromised promises of infrastructural access, as in many cases, local residents and civil society

organizations actually *want* these infrastructures (Brodie, 2020b; Brodie and Velkova, 2021; Burrell, 2020) in spite of their unsustainable resource use? Data scholars Segura and Waisbord (2019) caution against essentializing and romanticizing resistance, especially from the global south, in terms of data rights struggles, advocating instead for an analysis of datafication in specific contexts as a register of “contradictory forces: control and resistance, domination and citizenship, capitalism and human development” (p. 417). We can see, by contrasting advocacy for limiting digital growth with an acknowledgement of social discourses supporting for its expansion, that there is no easy solution to uneven growth when data systemic expansion is undergirded by intersecting logics of state developmentalism and the profit motives of multinational tech companies. At the very least, environmentally just development goals will not be achieved equitably under the infrastructural regimes of datafied accumulation, and in some cases communities are being turned into digital sacrifice zones through the expansion of extractive data infrastructure into new and already historically marginalized territories (see Childs, 2022).

Geographical approaches to the global expansion of capitalist infrastructures have long recognized the frontier making and breaking that comes along with resource making (Bridge, 2011; Rasmussen and Lund, 2018; Tsing, 2005: 27–50), which is ultimately an epistemological and spatial dynamic being reproduced within the growth of data infrastructures. This is not only an expansive characteristic, for example across the global north and south and “accumulation by dispossession” via land grabs by powerful multinational companies (Harvey, 2004; Thatcher et al., 2016), but one that represents the intensification of the extractive relationship into digital systems within the “metropole” itself (Couldry and Mejias, 2019a). The spatial character of environmental colonialism through digital infrastructure is not only representative of an epistemology of growth, but also its primary form of expression and intensification through infrastructural imbalances—which appear as lack of access and exclusion but are more complex and centripetal via forms of consumer integration via infrastructure.

Data colonialism’s infrastructural reach

The concept of “data colonialism” has been well-positioned as an analytic device for these dynamics of simultaneous aggressive integration through growth and infrastructural exclusion, or the intensification of “data relations” across individual data use and the political/spatial expansion of digital systems. Data has long been a tool of business and governance in an analog sense, but in the last few decades, as digital media scholar Srnicek (2016) demonstrates in *Platform Capitalism*, “the move to digital-based communications made recording exceedingly simple.

Massive new expanses of potential data were opened up, and new industries arose to extract these data and to use them so as to optimize production processes” (pp. 28–29). Data becomes a tool for more efficient production—including energy and the generation of value through environmental conditions (see Goldstein and Nost, 2022)—in its material and semiotic transformation into a *resource*.

Beyond infrastructure, access and use of digital data systems is also dependent upon a range of technological, and policy factors which span geographical (Kitchin, 2014), labor (Roberts, 2019), racial (Benjamin, 2019; Noble, 2018), and other modes of differential and discriminatory governance. Digital access, according to geographers Thatcher et al. (2016), acts to bring new users into a system of “data colonialism,” enacting a familiar geographical dynamic of coloniality—not only does data follow existing routes of colonial exchange, its dynamics of overaccumulation, investment, and dispossession occur through emergent regimes of digital data exchange and the tech companies that control its infrastructure. As Couldry and Mejias (2019b) articulate their take on the concept,

this new type of appropriation that works at every point in space where people or things are attached to today’s infrastructures of connection ... Just as historical colonialism over the long-run provided the essential preconditions for the emergence of industrial capitalism, so over time, we can expect that data colonialism will provide the preconditions for a new stage of capitalism that as yet we can barely imagine, but for which the appropriation of human life through data will be central.

To Couldry and Mejias, this surpasses Marxist theories of increasingly automated appropriation of labor as such (see Terranova, 2000)—it extends to the matter of physical and social life itself, and thus subjectivity. As a glossary definition expresses the core issues, resonating with scholarly advocacy of data sovereignty (Rodríguez-Lonebear, 2016) and data justice (Taylor, 2017), data colonialism is put simply “the process by which governments, non-governmental organizations and corporations claim ownership of and privatize the data that is produced by their users and citizens ... Producers of the data are the ones who are most often dispossessed of their right to own and control this resource” (Der et al., n.d.). Data colonialism thus describes intensive and extensive dynamics of compulsory integration and continued dispossession.

This is an obviously geographical *and* subjective dynamic, one with roots in biopolitical management and the privatization of governance as much as territorial resource control (although obviously these two spheres are intertwined). The centrality of data “dispossession,” which updates Harvey’s (2004) (by way of Marx’s) conception of accumulation *by* dispossession in neocolonial and

imperialist global systems, should speak to exactly this intersection of land and data relations. Big data is thus “not something new, but part of the continuous, entangled processes by which technologies operate under capitalism,” as Thatcher et al. (2016) contend. Social scientists Reardon and Tallbear’s (2012) analysis of data sovereignty and the biopolitics of colonial technoscience offers a more violent example of these entanglements between land and data in the racial construction of Indigenous land claims in the United States. As they demonstrate, the ongoing nature of data extraction and exploitation in and through colonized subjects represents a digitalized form of the ongoing histories of biopolitical management and dispossession of Indigenous peoples. As data scholar Rodriguez-Lonebear (2016) puts it, “The word ‘data’ comes from the Latin *datum*, meaning ‘something given.’ However, indigenous experiences under colonial control suggest that data more often means ‘something taken’” (p. 255). Struggles over data colonialism in these conceptions respond to the intensive/extensive dynamic of contemporary knowledge systems, focused on the intensification of governance and value relations through individual and collective subjects rather than *necessarily* geographical expansion (although this is certainly still involved). How does a settler colonial state, for example, maintain and expand territorial control via digital data or mapping? How does this entrench increased policing or militarization? These are key questions in the effects of pure data extraction on colonized communities or by neocolonial mechanisms on an unequal planet, and central to any analysis of the colonial relations of data.

But on its own terms, the theory of data colonialism reproduces a key misapprehension of global north-centered data discourse suggested above—that access to digital technology, and thus individual action on and through it, *defines* subjectivity to the degree that it is assumed to be co-constitutive of contemporary subjects. This is often framed as a holistic stream of “data relations” (Couldry and Mejias, 2019b), and takes the form of a totalizing digital apparatus that controls labor, production, circulation, and consumption. Data colonialism attends to existing and compounding imbalances *within* the global north and connected users and economies across global “digital divides” (see Appadurai, 2016), as defined as they are by class, race, gender, ability, citizenship, and location. However, subjective “intensification” of data relations via the feedback loops of digital media does not fully attend to the ways in which the differential impacts of *infrastructural expansion* remain key characteristics of digital growth—dynamics that Couldry and Mejias’ data colonialism describes but does not convincingly spatialize and contextualize, especially in its relationship to violence. This particular “one-worldism” of data discourse risks flattening the uneven resource logics of digital systems (see Milan and Treré, 2019), requiring a “skeptical, probing approach

to universalist generalizations drawn from cases in the North” to account for the actually existing granularities (and violence) of datafication in all contexts, north and south (Segura and Waisbord, 2019: 417).

Ricaurte (2019), in her critiques and expansions on the concept of data colonialism, encourages scholars to focus on the “bodies, affects, and territories” of those in the global south especially when understanding these subjective dynamics, whether that means how they access or how they are excluded from or violently subjugated by mechanisms of data extraction. As she argues, colonization writ large continues to reproduce “injustice within and across countries ... [and] erases alternate visions of the world so that technology can continue to operate as a renewed form of oppression” (p. 353). Through economy, political systems, knowledge, being and sensing, nature, and sociotechnical systems, data and its epistemic and material functions enact violence upon populations and territories through various mechanisms of collection, abstraction, analysis, and use (Ricaurte, 2019). There are many questions that can be answered by analyzing, for example, the dynamic interaction between discriminatory data representation and distribution and the political systems and social worlds that it shapes and acts within (see Chun, 2021), which are especially urgent in the apparent algorithmic automation of these systems via AI and machine-learning. Who “gets to be” a data subject in terms dictated by powerful digital companies and these proprietary systems? Whose digital lives and labors are commodified and made profitable without remuneration or consent? Who is policed? Surveilled? Excluded? Abandoned? But with these questions in mind, access to largely private data technologies, and the use of this data, cannot be reorientated toward more “just” ends without fundamentally confronting the infrastructural systems which are required to sustain digital access—a central backend of all fights for digital rights and “technology for good” is the uneven infrastructure itself.

We should be very aware that, for example, thinktanks and tech industry groups have incorporated “data colonialism” into their own frameworks, arguing that there is a “data gap” across the global north and south that can be fixed by more equitable data collection and analysis (see Dahmm and Moultrie, 2021). Reardon and Tallbear (2012) notice this compromised appropriation of data politics, seeing ideas of equitable data collection within a legacy of paternalistic practices that open the door for rampant misuse. These tendencies align under the guise of infrastructural development via big tech capital, who inherit the one-worldism and violent universality of colonial development tendencies enacted by private sector forces. But drawing on media scholars Powell et al.’s (2022) useful assertion, if “the ‘good’ in ‘technology for good’ continues to mask the structural issues inherent in our societies that cause the problems that need to be fixed through

technology” (p. 3), we need more “context-centered” approaches to counter this universalism. While data and access to it can be useful and redirected towards less harmful ends, the current administrations, software, and infrastructures by which it is collected, analyzed, and utilized are resolutely in the hands of states and corporations, from software to hardware. Bluntly, to quote digital humanities researcher Rodríguez-Ortega (2022), “We should not fool ourselves: the problem of the digital divide lies not only in the conditions of access but also ... the monopolistic control of the means and systems of production and distribution” (p. 109). This developmental impulse carried forward by private, digital data economies is instantiated not only by profit-driven data misuse and technologies of collection, but by the essentially infrastructural, energetic, and climatic supply chains of digital systems.

By paying attention to infrastructure, and specifically how the networks that power data centers map onto and intensify historical imbalances that *cannot be rectified through the data systems that they sustain*, planetary implications (and tactics) thus come into clearer focus. Infrastructures marshal and organize data relations as digital technologies capture information from the world and transform it into value. This is crucial in efficient modes of extraction and measuring—for example, in terms of Microsoft’s Terrain AI project in Ireland, which has seen the company partner with public funders to make a cloud-based platform to digitally map carbon storage and sequestration across different land uses (ostensibly for the measurement of future carbon accounting and crediting); or the datafied efficiency and optimization models developed by companies like Microsoft and Google to help the grid (and by extension enable their own facilities and technologies) to more flexibly interact with intermittent energy sources such as wind, solar, and tidal power (Radovanovic, 2020; Roach, 2022). The values and data in these projects are instantiated within and through digital infrastructures, as well as through the map of extractive activities that supply and sustain them, from energy systems to mining to multinational and outsourced manufacturing.

Data centers, in these contexts, become a tool of resource-making by processing and refining for different and intersecting extractive activities, as the final section will demonstrate. Researchers and activists have identified and unraveled these dynamics as they play out in colonial and postcolonial contexts, for example, through current landgrabs for (and struggles surrounding) multinational, hyperscale data infrastructures specifically in places like South Africa (see Pollio and Cirolia, 2022), Chile (Lehuedé, 2022b), the Arctic (Childs, 2022; Vonderau, 2019), India (Jha and Sengupta, 2022), and Asia Pacific (Hristova et al., 2023), or against extractive activities associated with “smart” and “green” technological development like lithium mining in the Atacama Plateau (Jerez et al.,

2021), wind development in Oaxaca, Mexico (Dunlap, 2017; Howe and Boyer, 2016), or solar farms and other green extractivist developments in northern Africa (Allan et al., 2021; Bruna, 2022). As the conclusion will argue, these environmental campaigns can be helpfully related to struggles defined by daily interactions with algorithms, screen media, social media content, and/or labor struggles connected to big tech’s dynamics of just-in-time production, outsourcing of content moderation, and AI labor (see Andrijasevic et al., 2021; Steinhoff, 2021). The extractive zones supporting “green” and “digital” industries, especially as big tech companies require more large-scale cloud infrastructures and energy to sustain them, are emerging spaces of data relations and data rights struggles.

This is not to unnecessarily expand the frame of data infrastructure to include all productive sites required to construct and supply data centers as singular infrastructures. Rather, this map of data infrastructural inputs required by “data colonialism” should demonstrate that the unevenness of data relations are *infrastructural* and *environmental* in their sustainment of uneven development in the form of large-scale industrial facilities like data centers. The expansion, scaling up, intensification, and institutionalization of data capture within the sociotechnical systems of the “fourth industrial revolution” manufactures the big data playing field on which critical infrastructure and environmental scholars and activists work today. In the final section, I will unpack the importance of data/energy relationships and environmental impacts in expanded data extractivism.

Energetic resources and data extractivism

In order for data to become a planetary “resource,” material and institutional networks must support it at global scales. Martín Arboleda, in his remarkable book *Planetary Mine*, articulates the ways in which the “fourth machine age” has reshaped global systems of extraction, and in fact expanded the footprint of resource extraction through data-driven and automated technologies. As Arboleda (2020) sees it, the study of discrete sites of extraction must necessarily be superseded by the concept of a “planetary mine” which “transcends the territoriality of extraction and wholly blends into the circulatory system of capital, which now traverses the entire geography of the earth” (p. 5). Arboleda argues that this is already laying a pathway to emergent planetary futures in both capitalist activity and anticapitalist movements, especially in terms of the concentration of extractive activities (like lithium mining) in Latin America. In short, data infrastructures’ expansion through software creates more and more hardware, needing both mineral inputs and waste management, which without radical restructuring means intensifying existing forms of exploitation and opening pathways to new ones at the planetary level. As instruments and

receivers of multiple forms and practices of extraction, data centers are thus imbricated within an array of intersecting activities at different sites, through different projects, rather than as undifferentiated big tech energy hogs.

In our article “New Extractive Frontiers in Ireland and the Moebius Strip of Wind/Data,” Bresnihan and I (2021) carve out what exact form these new and distributed extractive circuits are taking, especially looking at challenges faced during the simultaneous shift to renewable energy and the exponential growth of data centers in Ireland through the 2010s. To supply their energy needs and support their “green” transformation, tech companies are controlling more and more links along energy supply chains, and doing so through both investment and datafication. For example, Microsoft and GE partnering on a wind farm in Kerry in 2017 to supply energy for their data centers was not only public relations—it was designed as a testbed for the datafication of weather forecasting and the optimization of wind delivery and battery storage (Microsoft News Center, 2017), as “the atmosphere is thus drawn *into the cloud*” (Bresnihan and Brodie, 2021: 1647, emphasis original) through infrastructural partnerships. These forms of resource integration between data centers, energy companies, and the grid have intensified over the past several years with the use of Microsoft’s facilities to absorb excess wind capacity in their internal battery arrays (Roach, 2022).

These extractive relations are recognizable, and the processes of resource making and refinement can appear somewhat familiar from older energetic resources like coal and oil to wind and data (see Bridge, 2011; Knuth et al., 2022). But wind and other intermittent renewable energy are emerging as unique resources of and for datafication, presenting an opportunity to demonstrate that while resources are culturally and discursively constructed (Bridge, 2011; Kama, 2020), the entanglements of data and intermittent renewable energy are distinctive to a particular transformation of energy delivery that will see the intensified presence of big tech within evolving energy systems. Political ecologists and geographers have broadly theorized the ways that uneven development continue to shape the development of renewable energy as it replaces, at scale, carbon-intensive energetic resources, demonstrating the land grabbing and “green extractivism” that offer ostensibly decarbonized substitutes for the coal mines and oil wells of modernity (see Knuth et al., 2022; Voskoboynik and Andreucci, 2022). Data centers, in this assemblage, are both energetic *outputs* and infrastructures of energy transitions, meaning we need to be precise in how they are understood, imagined, and analyzed as essential industrial infrastructures for these expanding activities.

The resource semiotics and metaphors of “data extractivism,” including how data colonialism exploits data as a resource and dispossesses users of their data rights, have already fed ideas about the forms of extraction that arise through digital technologies (Couldry and Mejias, 2019a;

Thatcher et al., 2016). Platforms have “data extraction built into [their] DNA”, according to Srnicek (2016: 49), and these resource politics operate metaphorically through data discourse: “Simply put, we should consider data to be the raw material that must be extracted, and the activities of users to be the natural source of this raw material. Just like oil, data are a material to be extracted, refined, and used in a variety of ways” (p. 28). But as Couldry and Mejias note, “Extractive rationalities need to be naturalized or normalized, and, even more fundamentally, the flow of everyday life must be reconfigured and represented in a form that enables its capture *as data*” (2019b, emphasis original). Digital data cannot be fuel without infrastructure, and only by infrastructural naturalization can it emerge as a resource in itself. As geology and atmospheric science can do to oil and wind, data science then makes it legible as a resource, but one of its own self-perpetuating making through energy-intensive and mineral-intensive infrastructural conduits.

Thus, metaphorical claims and discursive naturalizations of data extractivism must be taken seriously, as Srnicek and others do (Taffel, 2023). Extractive industries conjure their worlds into being through these mechanisms, and that is exactly why and how they need to be contested. Material processes are necessarily imbricated within, and enabled by, political and economic narratives (Voskoboynik and Andreucci, 2022). These metaphorical and imaginative identifications, from the “cloud” to the social relations marshaled into data extraction and its accumulative processes, have material analogs and effects. So “symbolic” politics in data economies, including those described under the contested guise of data colonialism, with a close and critical analysis, help us recognize that data resources and infrastructures are nothing without supply chains (see Brodie, 2020a). Extraction itself has never been a geographically discrete activity—it has *always* relied on networked, multi-scalar, and often conflicting technoscientific discourses and geographical relations in some way. Crawford (2021) warns that

The field of AI is explicitly attempting to capture the planet in a computationally legible form. This is not a metaphor so much as the industry’s direct ambition. The AI industry is making and normalizing its own proprietary maps, as a centralized God’s-eye view of human movement, communication, and labor.

And this is not only related to the more “infrastructural” approach, as she unravels in detail. These are industries that desire to be “the dominant way of seeing. This colonizing impulse centralizes power in the AI field: it determines how the world is measured and defined while simultaneously denying that this is an inherently political activity” (Crawford, 2021).

The infrastructural production of data as a *resource* drawn from networked activities, and the functional requirements of energy-intensive resource inputs for the operation of its resource making, means that the dismantling of metaphorical politics is essential to critical engagement with data extractivism. While theorists like Ricaurte (2019) and Sadowski (2019) place infrastructure as one (albeit major) layer of a multiply extractive apparatus of data capitalism, in the form and function of data centers, I argue that we have a much more centralized version and site of analysis for these various inputs—through which many different infrastructures meet in the multiply extractive facility of a data center, which essentially refines data as a resource via extremely resource intensive processes drawing from a number of public utilities, climate and land resources, private contracts, consumer data collection, and labor pools. As energy humanist Kinder (2023) evokes in the context of multinational data centers in Québec drawing hydropower generated on and via Indigenous lands and waters across the province, “Beyond metaphor, settler-colonial relations fuel the data center industrial complex and its adjacent initiatives like cryptocurrency. Here, the deepened impacts on land and territory cannot be overlooked as they operate as another site of ambivalence and intensity” (p. 107). Drawing on data infrastructure scholar Mél Hogan’s (2021) notion of the “data center industrial complex” to theorize what he calls “platform energetics,” or the expanded mediation between users and resources via platforms, Kinder (2023) emphasizes that data centers thus marshal energetic and extractive relationships to land in a concentrated infrastructural conduit as material and semiotic refiners of data, which in turn power data capitalism. The circuitry of these “platform energetics” can be illustrated not only through the many relations *mediated* through data/energy, but also through the world-eating vastness of global supply chain management within planetary computation – not just Arboleda’s (2020) “planetary mine”, but the interoperational ability of extractive industries, from multinational digital corporations to Canadian mining companies speculating and operating from Ireland to Chile. As critical social science and humanities scholars, our frameworks must account for both the new and familiar shapes the inequalities of extractivism take at a planetary scale that also accounts for the apparently irresolvable contradictions and imbalances of global connectivity as managed by multinational tech companies that occur at localized sites of data infrastructural encounter.

To combat the legacies of environmental colonialism in contemporary data/energy/extractive frameworks, data infrastructure studies can uniquely push against the reification of data as socially representative in its consumer and state functions by stubbornly staying with these physical, technical systems within emerging territorial formations and planetary relations of data. While this may be crassly

geographical to some viewpoints, data extractivism occurs *spatially* and rakes across unequal and colonially defined global territories and spatial relations. As anthropologist Spice (2018) argues, in alignment with studies of Indigenous data justice and sovereignty, the “production” of critical infrastructure describes a degenerative process by which Indigenous life is destroyed “to make way for capitalist expansion” (p. 41). While undoubtedly reproducing epistemologies also embedded in physical operations, these are not necessarily machinic—they can be seen in government, corporate development, community organizing, and most other spheres of activity embedded within ongoing global inequalities. In following Crawford’s (2021) holistic approach, then, I want to reiterate through this article that when “calls for data protection, labor rights, climate justice, and racial equity [are] heard together ... different conceptions of planetary politics become possible”—a data infrastructure studies *for* an unequal planet, toward a truly planetary environmental justice.

Conclusion

An approach to data infrastructure at its various sites of expansion and development expands our understanding of the material, lived implications of digital technologies beyond how they are used, consumed, and optimized. This is not to supersede sociological and representational approaches to big data, algorithms, and social media platforms as such. Rather, it is to demonstrate how data extraction, algorithmic injustices, and the maintenance of inequality through these consumer-based technologies can be related to anti-imperialist and environmental justice goals through a focus on their infrastructural footprint. How do algorithmic injustices relate to lithium extraction in the Atacama Plateau? How does a boggy riverside land plot in Leitrim relate to a scenic tourist beach in Mayo? What about an industrial site off the M50 motorway outside of Dublin, with seafloor ecosystems several kms offshore in the Irish Sea? As these examples elicit, struggles against, for example, data mining operations, or a multinational battery farm, or a gas-fired data center, are more related to mineral mining companies and fiber optic cable landings and offshore wind than they initially appear. Activating and understanding infrastructural footprints of these connections is essential for a planetary anti-extractivism that connects data rights with data infrastructural struggles without resorting to individualized and consumer-centered solutions to digital sustainability.

Environmental scholar Buck (2022) argues that, in relation to potential overlaps between digital activism and net zero goals in avoiding what she calls “platform determinism,”

[environmental activists] may be actively opposed to technological interventions, or understandably dismiss net

zero as a narrow, technocratic goal. Meanwhile, people with expertise in algorithmic justice issues might not be tracking developments in the environmental sphere ... But each of these communities has critical things to contribute.

As sociologist Vgontzas (2022) also describes the work of Amazon union leaders and activists aligning with environmentalists, a coalitional approach is needed—and this includes across academia, policy, and labor as much as alignments between diverse digital rights advocates, environmental activists, anti-extractivist, and anticolonial movements with shared objectives towards reversing or disrupting the trends of big tech's growing co-optation of and imbrication within global environmental objectives through their content, analysis, political, and infrastructural apparatuses. Critical environmental data scholar Sebastián Lehuedé writes about the Contested Data Territories initiative (2022a), co-initiated with environmental media researchers Julia Rone and Hunter Vaughan in 2021, that:

Data centre activism shares some of the goals of other groups such as the digital rights movement and technology companies' unions. Perhaps, the key for democratising technology lies in the capacity of these groups to unite their forces in solidarity against a common adversary: a handful of increasingly powerful and rich technology companies undermining privacy, democracy and the environment at a planetary scale.

Virtually bringing together people speaking in contexts from Chile, to Ireland, to the Netherlands, each context hosting (and communities battling) big tech hyperscalers Google, Microsoft, and Amazon, this event modeled one (of many) way(s) forward through a wide-ranging, coalitional approach that united anti-extractivist and environmental justice groups fighting powerful forces in very different global contexts, sharing resources and tactics toward more just data futures.¹ As Rone (2023a) similarly concludes in a 2023 provocation about cultivating emerging networks of data center activism, the building of alliances is necessary, but with a crucial recognition in place about the importance of embedded, careful, and context-specific research to craft solidarities and alignments between academics and activism. Expanding initiatives such as this will be crucial in data battles to come.

The unevenness and imbalances of digital networks, especially in terms of the extension of digital systems into energy and other environmental resources and management, means that attention to activism and multifaceted conceptual frameworks across spatial and epistemological practices are required to keep each of these sites and issues in mind, even as the planetary scales of big tech, of climate change, threaten to overwhelm the frame. Solutions which posit growth through optimization and smarter, greener economies inevitably lead down familiar

pathways of imbalanced infrastructural relations. It may seem obvious to ask at this point, but is the expanded extraction of energetic resources including lithium, wind, solar, water, or “critical minerals” and resources from “underdeveloped” (overexploited) communities worth the moral costs of ongoing connection? What would “green development” focused on the thriving and poverty alleviation of resource-rich places in the global south look like beyond resource nationalism? Beyond paternalism and developmentalism? Where communities thrive, and responsibilities, costs, and impacts shift equitably, without simply onshoring extraction to the global north (Riofrancos, 2022)? And how do you attend to the differentiation within and across territories in ways that contribute productively to projects advocating and struggling for more just systems? Data infrastructural systems need radical democratic restructuring and redistribution of responsibility which actively centers the voices and experiences of those most negatively impacted by the ongoing and expanded inequalities of material extraction during current environmental transitions. But to craft these more just and careful systems, we need to first unsubscribe from data capitalism's limited planetary futures.

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Note

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